Mohamed H. Essa

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SUMMARY

Cloud / DevOps Engineer with 2+ years of production experience building CI/CD pipelines, container platforms, and Terraform-managed AWS infrastructure. AWS Solutions Architect Associate (SAA-C03) certified; CKA and AWS DevOps Pro in progress. Proven track record cutting release cycles, eliminating manual toil, and standing up first-of-kind IaC and observability practices on engineering teams.

TECHNICAL SKILLS

Cloud (AWS): EC2, EKS, ECS Fargate, Lambda, RDS Aurora, S3, CloudFront, Route 53, IAM, VPC, CloudWatch, GuardDuty, Security Hub
IaC: Terraform, Terraform Cloud, AWS CDK, CloudFormation
Containers & Orchestration: Docker, Kubernetes (EKS, k3s, kind, self-managed), Helm, ArgoCD, Kustomize
CI/CD: GitHub Actions, GitLab CI, Jenkins, ArgoCD, Trivy, Syft, cosign, Kyverno, SonarQube
Observability: Prometheus, Grafana, ELK, Datadog APM, PagerDuty, CloudWatch Logs Insights
Languages & Scripting: Python (Boto3, FastAPI), Go, Bash, Node.js; REST, GraphQL, OpenAPI
Databases: PostgreSQL, MySQL, Redis, MongoDB, DynamoDB
Networking & Linux: VPC design, ALB/NLB, Transit Gateway, VPN; Ubuntu, Amazon Linux, RHEL, systemd

EXPERIENCE

DevOps & Platform Engineer Aug 2022 – Jul 2024
Intrazero — Digital Product Studio · Cairo, Egypt

- Built and owned end-to-end **GitHub Actions CI/CD pipelines** for 4 production applications — automated builds, test gating, code signing, and store delivery; **cut release cycle from 3 days to same-day** with zero manual steps.
- Provisioned AWS environments (EC2, RDS, S3, CloudFront) with **Terraform**; introduced remote state via S3 + DynamoDB locking and separated staging/production workspaces — first IaC practice on the team, replacing click-ops across 30+ resources.
- Containerised backend services (Node.js, FastAPI) with Docker and authored **Helm charts** for an EKS cluster, including namespace-level RBAC, resource quotas, and rolling-update policies; reduced production deploy time from ~45 minutes to under 4.
- Stood up **Prometheus + Grafana** monitoring with p50/p95/p99 SLO dashboards and PagerDuty routing — gave the team its first structured on-call process and **cut mean time to detect by ~60%**.
- Drove adoption of an OpenAPI-first contract workflow across mobile and backend teams; integration defects **dropped 35%** in the first quarter and new-engineer onboarding fell from 4 days to 1.

PROJECTS

Slipway · *Kubernetes (kind), Terraform, Argo CD, Kyverno, cosign, Prometheus, Go*

- **GitOps platform on a laptop**: *make up* builds a 3-node kind cluster (**Terraform**); **Argo CD** installs Envoy Gateway, Prometheus/Grafana/Loki/Tempo and CloudNativePG from Git; EKS version validated offline.
- **Signed supply chain**: Trivy scan, Syft SBOM, cosign signature; **Kyverno admits only signed images**.
- **SLOs with burn-rate alerts**; **chaos day under k6 load**: **0 failed requests** in releases; a node loss exposed single points of failure (48% errors for 6 min), fixed the same day.

Serverless URL Shortener · *AWS Lambda, API Gateway, DynamoDB, Python, CDK, WAF*

- **Sub-20ms p99 latency at 10,000+ requests/day** with zero ops overhead. Fully serverless, deployed via AWS CDK with a custom domain, WAF rate-limiting, and CloudWatch alarms; DynamoDB single-table design **keeps monthly cost under $1**.

Containerised API Platform · *Docker, Kubernetes (k3s), Helm, FastAPI, PostgreSQL, GitHub Actions*

- Self-hosted platform serving 3 internal APIs on a k3s VPS cluster — Helm charts per service, GitHub Actions builds and pushes images to GHCR on merge, automatic rolling deploys. **Total infra cost: $18/month**; primary hands-on environment for CKA preparation.

Dev-Container CLI · *Go, Docker SDK, Bash*

- CLI tool that provisions isolated dev containers for Python, Node.js, and Go projects in under 5 seconds, with specs version-controlled alongside each repo. **Eliminated environment drift across a 15-person team** — zero “works on my machine” incidents since rollout.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) · **AWS Certified DevOps Engineer – Professional (DOP-C02)** — In Progress · **Certified Kubernetes Administrator (CKA)** — In Progress

EDUCATION

B.Sc. Computer Engineering, *Benha University — Cairo* Sep 2019 – Jun 2024